

CSE490(D): Football Data & Analytics

Sample Questions (Lectures 9, 10, 11,12)

TAV

1. Definition / Short Answer

1. What is pitch control in football analytics?
2. What is a Voronoi diagram?
3. Define internal load in sports science.
4. Define external load and give two examples.
5. What is meant by time-to-intercept?
6. What does the parameter λ represent in pass probability modeling?
7. State one limitation of tracking data.
8. What is sports science?

2. Analytical Questions

1. Explain why event data alone is insufficient for analyzing off-ball player behavior.
2. Explain the concept of passing triangles and their importance in football. (Read suggested book chapter)
3. Why are defenders advised not to stay on Voronoi boundaries?
4. Compare the following approaches:
 - Time-to-intercept model
 - Machine learning model

Focus on interpretability, flexibility, and realism.

5. Explain the roles of:

- Time-to-intercept
- Time-to-control

Why are both necessary in pass probability modeling?

6. Explain how simple local rules lead to complex global behavior in football movement.
7. Why does the pass probability model ignore the Magnus force? What are the implications?
8. Why is pitch control modeled as a probability surface instead of discrete regions?
9. Explain advantages and disadvantages of different external and internal load frameworks.
10. Is being in underload condition helpful for the players' fitness? Explain.
11. What is physiological and mechanical reserve? How they get depleted? How to monitor and recover these reserves?
12. Compare modern tracking technologies? (Computer vision & optical tracking, AI-powered video analysis, wearable + optical fusion)
13. What are the forces that control ball trajectory? Why pass probability model only addresses pass that travel in straight line.

3. Numerical Problems

1. A ball travels 30 meters at a speed of 10 m/s.
 - Calculate the arrival time.
 - Explain how this value is used in pitch control modeling.
2. A pass travels 21 meters with a speed of 15 m/s.
 - Compute the ball arrival time.
 - Player X reaches in 1.2 s and Player Y in 1.5 s.
 - Who has a higher probability of intercepting the ball? Explain.
3. A player stays near the ball for:
 - 0.1 seconds
 - 0.5 seconds

Which case has a higher probability of control? Explain using intuition from the exponential model.

4. Consider a pitch scenario:
 - Attacker A is 5 m away from a location.
 - Defender B is 7 m away but moving faster.
 - Who is more likely to control the ball?
 - Explain using time-to-intercept and probability concepts.

4. Code Interpretation

```
def arrival_time ( distance , speed ) :  
    return distance / speed  
t_ball = arrival_time (20 , 10)  
t_player = arrival_time (20 , 8)  
if t_player < t_ball :  
    print ( " Player controls ball " )  
else :  
    print ( " Ball arrives first " )
```

1. What does this code compute?
2. What assumption does it make about player movement?
3. Why is this model unrealistic in real football analytics?

5. Higher Thinking Ability

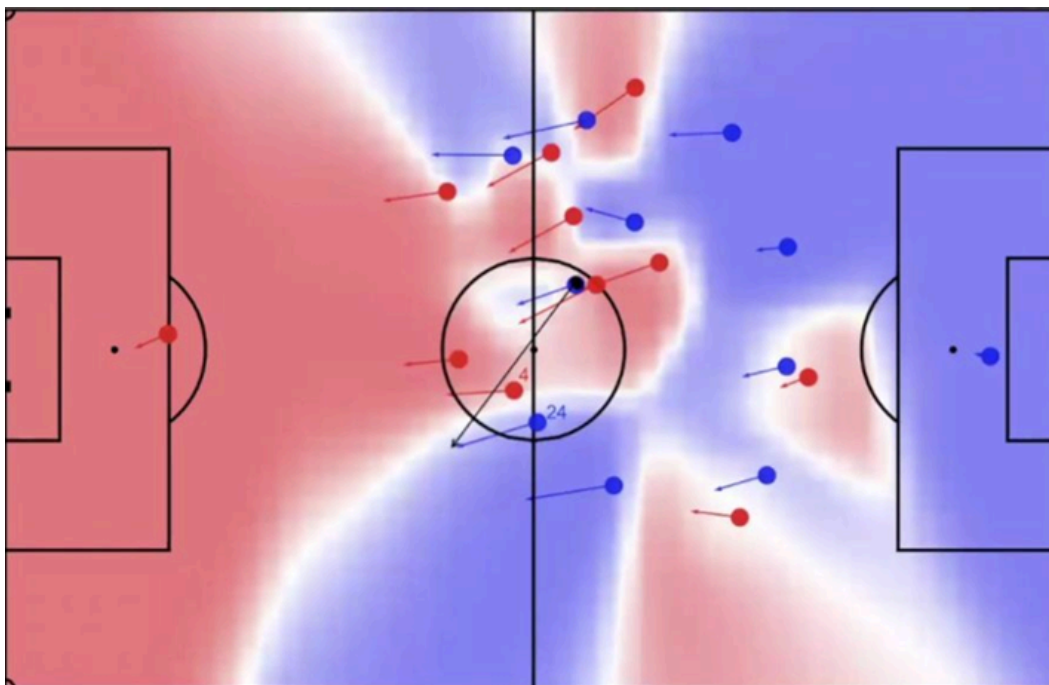
1. Explain why pitch control models are probabilistic rather than deterministic.
2. How can pass probability models be used for:
 - Tactical analysis
 - Player evaluation
3. Explain the concept of uncertainty in interception modeling and its importance.
4. In a pitch control map, a defender is positioned far from his natural position (e.g., left-back appearing on the right side).
 - Why might this happen?
 - Explain using the concept of “defense in attack”.

5. Given a pitch visualization:

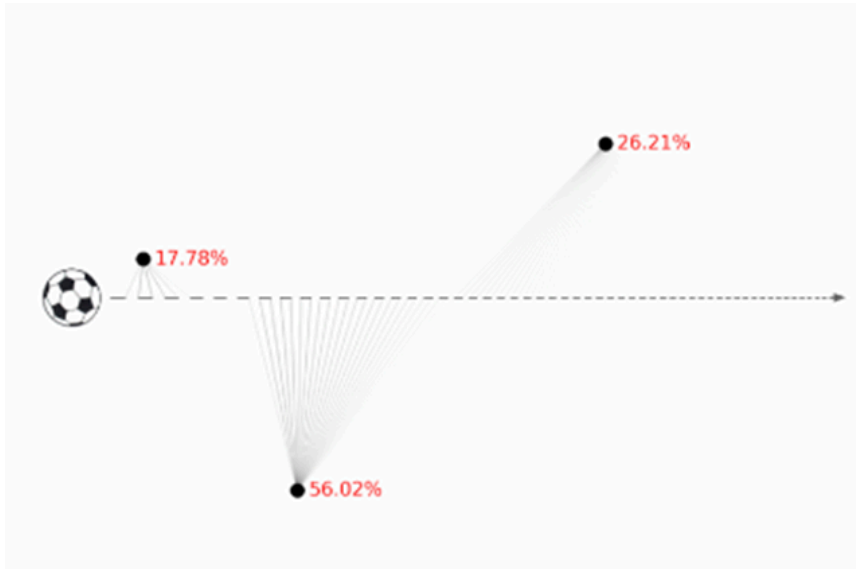


- Is it a good offside trap? Explain your answer.
- How can defenders maintain offside trap?

6. Comment on the quality of the pass in the picture based on pitch control model.



7. What can you say about the probability of the players to receive the ball?



8. How can you combine Expected Possession Value (EPV) and Pitch control to create a better metrics to rate a pass? (**Refer to Spring 26 final Question 5**)

9. What does the following code block do? Explain the process. How can it improve the xG modeling?

```
def players_in_triangle(test_shot, track_df):
    #get id of the shot to search for tracking data using this index
    test_shot_id = test_shot["id"]
    #get all opposition's player Location
    player_position = track_df.loc[track_df["id"] == test_shot_id].loc[track_df["teammate"] ==
False][["x", "y"]]
    #checking if point inside a triangle
    x1 = 105
    y1 = 34 - 7.32/2
    x2 = 105
    y2 = 34 + 7.32/2
    x3 = test_shot["x"]
    y3 = test_shot["y"]
    xp = player_position["x"]
    yp = player_position["y"]
    c1 = (x2-x1)*(yp-y1)-(y2-y1)*(xp-x1)
    c2 = (x3-x2)*(yp-y2)-(y3-y2)*(xp-x2)
    c3 = (x1-x3)*(yp-y3)-(y1-y3)*(xp-x3)
    #get number of players inside a triangle
    return len(player_position.loc[((c1<0) & (c2<0) & (c3<0)) | ((c1>0) & (c2>0) & (c3>0))])
```